

AWS CLOUD (TICKET RESERVATION SYSTEM)

AWS

AWS stands for Amazon Web Services. It is a cloud computing platform provided by Amazon. It helps us run applications, store data and manage servers over the internet instead of using our own computer. AWS is used because it is reliable, secure and we only pay for the services that we use. It can also handle many users at the same time without slowing down.

AWS Cloud

Cloud computing means storing files, databases and applications on the internet instead of keeping everything on a local computer. It allows users to access the system from anywhere using an internet connection. It also makes backup and recovery easier if something goes wrong.

Amazon EC2 (Elastic Compute Cloud)

Amazon EC2 is a virtual server provided by AWS. It is used to host websites and applications. We can increase or decrease the server size depending on the number of users. This makes it useful for projects where the number of visitors changes.

Use in Ticket Reservation System

- I can host my ticket reservation website on EC2.
- It can handle many users booking tickets at the same time.

Amazon S3 (Simple Storage Service)

Amazon S3 is a storage service used to store files online. It is secure and can store a large amount of data. The files can be accessed whenever they are needed.

I can store:

- Event posters
- User profile pictures
- PDF tickets

Use in Ticket Reservation System

Instead of saving ticket files and images on my computer, I can store them in S3. Users can download their tickets whenever they want. If a user loses the ticket, it can be downloaded again from the system.

Backup Files

Taking backups is important because data can be lost due to software errors or accidental deletion. AWS allows automatic backup of important files.

Use in Ticket Reservation System

I can keep a backup of booking details, customer information and ticket records in AWS. If the main system stops working, I can restore the data from the backup. This helps prevent data loss.

AWS Lambda

AWS Lambda is a serverless service. It runs code only when an event happens. There is no need to keep a server running all the time.

Use in Ticket Reservation System

- Send booking confirmation emails automatically.
- Generate PDF tickets after payment.
- Update seat availability after a booking.

AWS Security

AWS provides different security features to protect data and applications.

Some security features are:

- User Login
- Password Protection
- Data Encryption
- Access Control

Use in Ticket Reservation System

Only authorized users can access the system. Customer information, payment details and booking records remain secure. Different users can have different permissions based on their role.

AWS IAM (Identity and Access Management)

AWS IAM is used to control who can access AWS services and what they are allowed to do. It helps keep the system secure by giving different permissions to different users.

Use in Ticket Reservation System

Admin

- Full access to AWS and the ticket reservation system.
- Can add or remove events.

Event Manager

- Can create events and update seat availability.
- Can check booking reports.

- Cannot change AWS account settings.

Customer

- Can register and log in.
- Can search events.
- Can book or cancel their own tickets.
- Can download booked tickets.

Amazon RDS (Relational Database Service)

Amazon RDS is a managed database service provided by AWS. It is used to store application data such as customer details, bookings and payment records. AWS manages backups, updates and maintenance.

Use in Ticket Reservation System

- Store customer information.
- Store booking history.
- Store event details.
- Keep payment records safely.
- Retrieve booking information quickly whenever needed.

Amazon CloudWatch

Amazon CloudWatch is used to monitor AWS resources and applications. It helps detect problems and shows system performance.

Use in Ticket Reservation System

I can monitor server performance, CPU usage and website traffic. If the server becomes overloaded or stops working, CloudWatch can send alerts so the issue can be fixed quickly.